

Tutorial audit: data-fitted prediction of phi2-dependent output amplitude

Purpose

This report reproduces, from the saved CSV artifacts, the reported numerical comparison between a data-fitted hybrid-MZI power model and a later, high-visibility phi2-fringe map.

Reported aggregate result

	predicted	measured	MAE
PD contrast mean	0.571	0.585	0.027
PAX ptotal mean	0.619	0.657	0.041

What is, and is not, being claimed

The result demonstrates that the calibrated effective-analyzer model predicts the DEGREE of phi2 power modulation in a later high-contrast state to a few hundredths of contrast. It does not establish a unique physical source for the nonideal power modulation, nor is it a blind prediction of fringe phase. The current phil branch alpha is selected from the fringe-map sine quadrature; that conditioning is shown explicitly on page 5.

All values in this report are computed from files on the next page. No fitted coefficient or reported contrast was entered manually into this document.

1. Evidence ledger: exact input data and derived artifacts

Raw field-model calibration

```
-----  
experiments/polarization_locking/2026-08-13/192530_field-model-calibration_phi2-power-model-  
rows = 984; SHA-256 prefix = 70aa7cfe62be8823  
phi1 RP commands = 0.00000, 0.18074, 0.36148, 0.54222 V  
phi2 RP commands = 0.000 ... 0.200 V (41 points)  
conditions = path A only, path B only, both paths  
two passes: forward and reverse
```

Raw high-contrast phil fringe map

```
-----  
experiments/polarization_locking/2026-08-13/200934_phil-fringe-map_phil-fringe-map-1/data.cs  
rows = 714; SHA-256 prefix = e8719481ab444339  
phi1 RP commands = 17 points, 0.000 ... 0.722963 V  
phi2 RP commands = 21 points, 0.000 ... 0.200 V  
both paths open; phil forward and reverse passes
```

Derived files used below

```
-----  
analyzer coefficients: experiments/polarization_locking/2026-08-13/192530_field-model-calibr  
SHA-256 prefix = 3c803a9e5ea4ecad  
fringe fits: experiments/polarization_locking/2026-08-13/200934_phil-fringe-map_phil-fringe-  
SHA-256 prefix = 6ca85ad6af011135  
comparison rows: experiments/polarization_locking/2026-08-13/200934_phil-fringe-map_phil-fri  
SHA-256 prefix = b59266a8c88eb988
```

2. From the ideal propagated field to an effective detector model

Ideal equal-amplitude field propagation at the selected final port gives

$$S(\phi_1, \phi_2) = \begin{bmatrix} \cos(\phi_2), \\ \sin(\phi_2) \cos(\alpha), \\ \sin(\phi_2) \sin(\alpha) \end{bmatrix}^T,$$

where α is the effective azimuthal phase of the ϕ_1 path, including the static phase convention / offset. In the ideal static model $\alpha = \phi_1 + \delta$. The experiment shows $\alpha(V_{RP})$ is history-dependent, so α is kept as an empirical branch coordinate rather than forced to be linear in RP voltage.

For each instrument d , the smallest identifiable first-order power model is

$$\begin{aligned} P_d &= B_d + h_d \text{ dot } S \\ &= B_d + h_{1,d} \cos(\phi_2) \\ &\quad + [h_{2,d} \cos(\alpha) + h_{3,d} \sin(\alpha)] \sin(\phi_2). \end{aligned}$$

B_d is that detector/condition's mean native instrument level. h_1, h_2, h_3 are fitted effective analyzer coefficients, in the same native units. This is a linear response to Stokes state—not an assertion that the PAX or PD is an ideal polarization analyzer. It is the most constrained model supported by the present two-axis calibration.

At fixed α this has the ordinary fringe form

$$\begin{aligned} P_d &= B_d + C_d \cos(\phi_2) + D_d \sin(\phi_2), \\ C_d &= h_{1,d}, \\ D_d &= h_{2,d} \cos(\alpha) + h_{3,d} \sin(\alpha). \end{aligned}$$

Thus the predicted modulation amplitude is $A_d = \sqrt{C_d^2 + D_d^2}$, and the dimensionless predicted contrast is A_d/B_d . The PD's OD 2 filter changes its absolute level but cancels from contrast; PD and PAX power units are never equated in this analysis.

3. Coefficients actually fitted from the A/B/both calibration

detector / both paths	B	h1	h2	h3	R2	fit RMS
PD (V)	0.24333	0.13204	0.010505	0.042949	0.900	0.032233
PAX ptotal	0.00023116	-0.00013261	-9.5971e-06	-5.3563e-05	0.793	5.0424e-05

Each row is the least-squares solution of $P=B+h_1*S_1+h_2*S_2+h_3*S_3$ over 328 both-path raw samples (four ϕ_1 biases $\times$ 41 ϕ_2 commands $\times$ two passes). These are the coefficients used to transfer the amplitude prediction.

4. Fringe map: measured contrast and empirical alpha

At every fixed phil RP command and scan direction, the raw both-path map is independently fit for each detector to

$$P_{\text{meas}}(\phi_2) = b + c \cos(\phi_2) + d \sin(\phi_2).$$

The reported measured fringe quantities are

$$\begin{aligned} A_{\text{meas}} &= \sqrt{c^2 + d^2}, \\ \text{contrast}_{\text{meas}} &= A_{\text{meas}}/b, \\ \psi &= \text{atan2}(-d, c). \end{aligned}$$

Equivalently, the normalized measured quadratures are

$$\begin{aligned} c/b &= \text{contrast}_{\text{meas}} \cos(\psi), \\ d/b &= -\text{contrast}_{\text{meas}} \sin(\psi). \end{aligned}$$

Why alpha is empirical

The later map showed that phil is not a single reversible linear phase scale. For each phil command and direction, the comparison code chooses alpha on a dense $[0, 2\pi)$ grid by minimizing, jointly for PD and PAX,

$$\text{sum}_d \{ [h_2, d \cos(\alpha) + h_3, d \sin(\alpha)] / B_d - (d/b)_{\text{meas}, d} \}^2.$$

This uses the map's sine quadrature (fringe phase information) to choose the current optical branch. It does NOT use the measured amplitude/contrast as an input to the amplitude calculation. Once alpha is selected, $C=h_1$ and $D(\alpha)$ are inserted into $\sqrt{C^2 + D^2}/B$ to make the predicted contrast.

Consequently: the contrast comparison is a conditional transfer validation, not a blind future-state forecast. A true blind phase-and-amplitude forecast requires a state-matched, reproducible $\alpha(V_{\text{RP}})$ calibration.

5. Worked numerical example: forward $\phi_1 = 0$ RP V

detector	alpha	C/B=h1/B	D/B	predicted A/B	measured A/b	error
PD	0.032044	0.54263	0.048803	0.54482	0.54571	-0.00089288
PAX	0.032044	-0.5737	-0.04892	0.57578	0.58784	-0.012054

Example calculation: PD uses $B=0.2433345$, $h_1=0.1320399$, $h_2=0.0105049$, $h_3=0.0429494$ and $\alpha=0.032044$ rad. Therefore $C/B=0.5426272$, $D/B=0.0488033$, and $\sqrt{(C/B)^2+(D/B)^2}=0.5448174$. The stored measured PD contrast is 0.5457103.

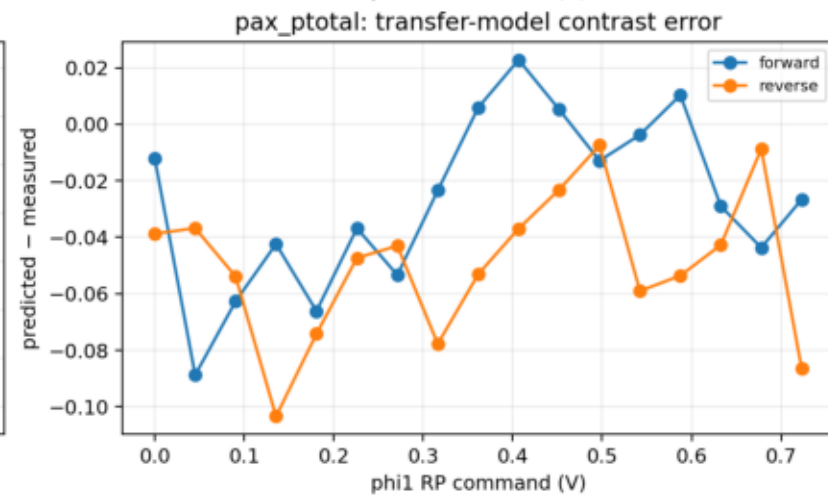
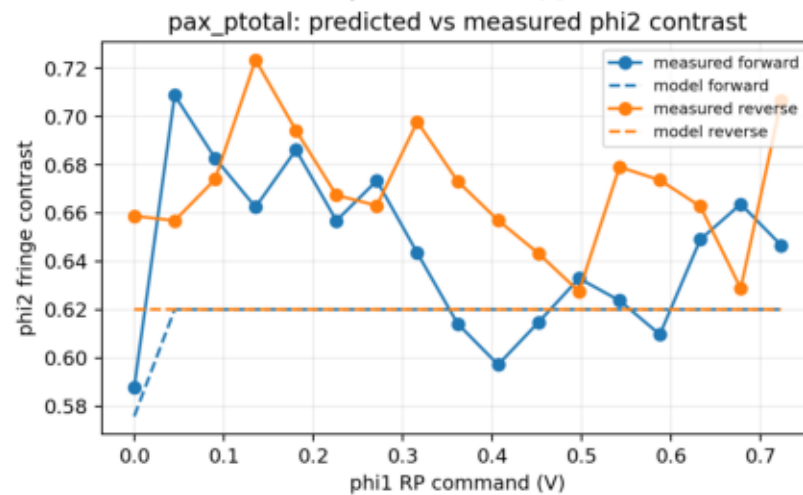
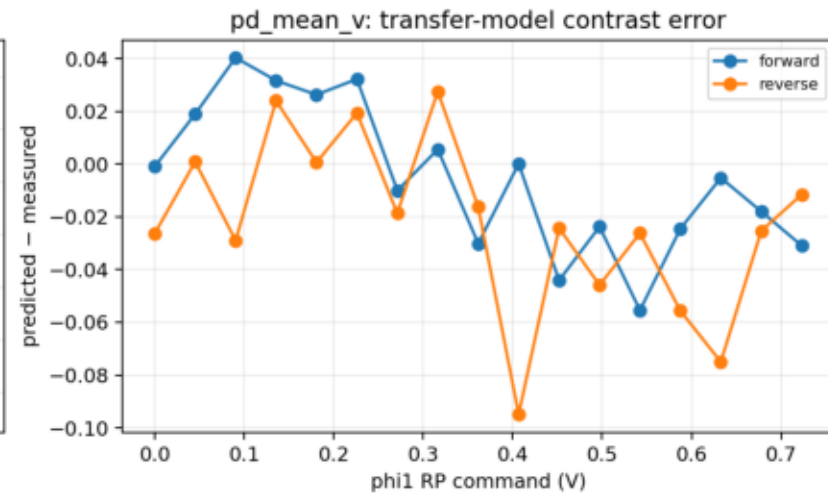
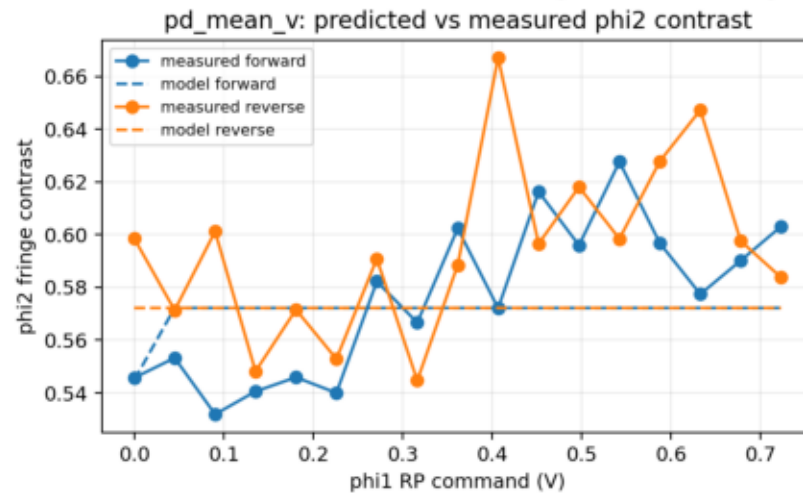
6. Aggregate comparison: all phi1 commands and both scan directions

output	N	mean predicted	mean measured	MAE	RMSE	bias	median rel. error
PD	34	0.571	0.585	0.027	0.034	-0.014	4.3%
PAX ptotal	34	0.619	0.657	0.041	0.048	-0.038	6.1%

N=34 for each detector: 17 phi1 commands x forward/reverse. MAE is mean |predicted contrast – measured contrast|. The requested 0.571/0.585/0.027 and 0.619/0.657/0.041 values are rounded directly from this table's underlying rows.

7. Stored numerical comparison plot

Field-model prediction of phi2-dependent output amplitude



8. Interpretation, checks, and next measurement

What the close amplitude agreement supports

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1. A substantial ϕ_2 -dependent output-power term is present and can be represented quantitatively by the fitted effective analyzer vector.
 2. The dominant calibrated modulation scale transfers across the two runs: PD MAE=0.027 contrast and PAX MAE=0.041 contrast.
 3. Agreement in both final-output instruments makes a single-instrument readout artifact less plausible as the sole explanation.

What it does NOT prove

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1. It does not uniquely identify the physical origin: nonideal splitting, polarization leakage, path-dependent loss, mode overlap, and detector response can produce effective h coefficients.
 2. It does not prove a stable ϕ_1 voltage-to-phase transfer. The alpha branch was conditioned on the later map's sine quadrature precisely because the measured forward/reverse ϕ_1 behavior was not reproducible.
 3. The calibration and high-contrast map were separate optical states. The joint-alpha residual in the comparison CSV measures this imperfect transfer.

Strong next test

Acquire field-model-calibration and ϕ_1 -fringe-map back-to-back while high contrast is maintained, then run `validate_phi2_power_model.py`. A state-matched repeat that preserves the present low MAE AND improves quadrature/phase transfer would be strong evidence that this is a stable predictive model.